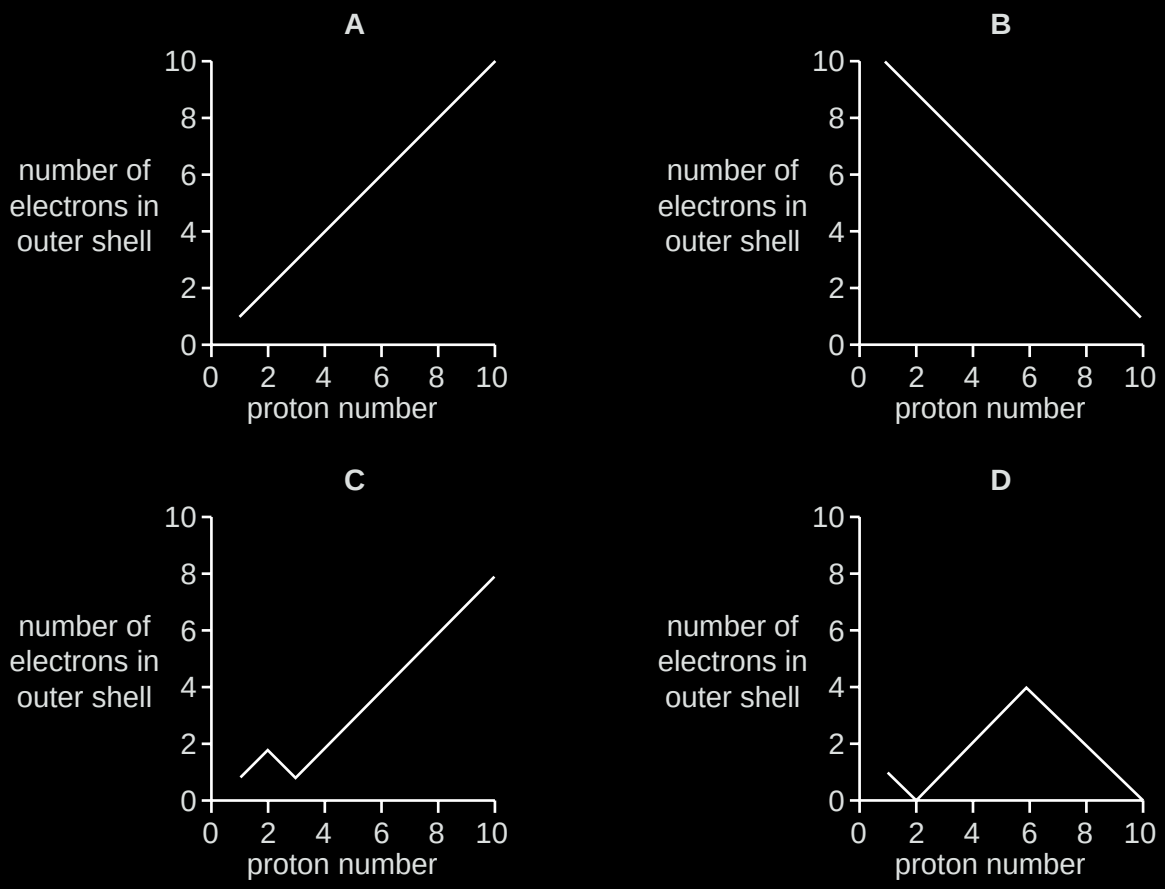


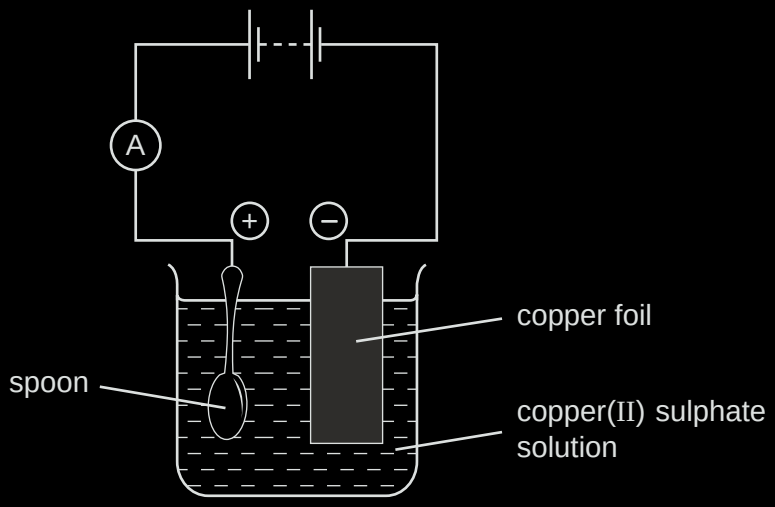
7 The symbols and electronic structures for some elements are shown below.

silicon, Si (2,8,4) oxygen, O (2,6) hydrogen, H (1)

fluorine, F (2,7) nitrogen, N (2,5)

Which formula is correct for a compound containing silicon?

A Si_4F **B** SiH_4 **C** SiN_5 **D** Si_2O



The experiment did **not** work.

What was the mistake in the apparatus?

A A variable resistor should be included in the electrical circuit.

B Dilute sulphuric acid should be used as the electrolyte.

C The copper electrode should all be in the solution.

D The spoon should be the negative electrode.

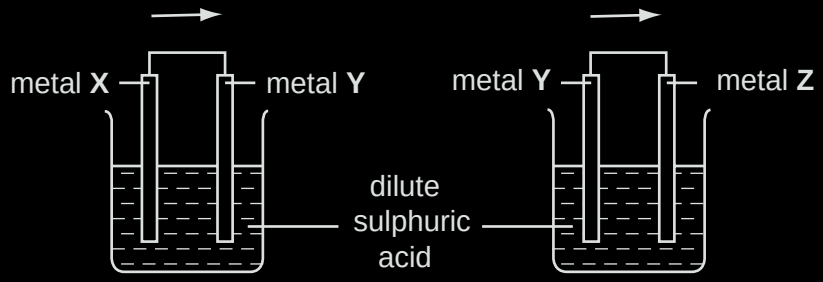
13 Which pair of substances act as reducing agents in the blast furnace?

A carbon and oxygen

B carbon monoxide and carbon dioxide

C carbon and carbon monoxide

D carbon dioxide and oxygen



Which set of metals would give the electron flows in the directions shown?

	metal **X**	metal **Y**	metal **Z**
A	Ag	Cu	Zn
B	Ag	Zn	Cu
C	Cu	Zn	Ag
D	Zn	Cu	Ag

15 The equation below shows an exothermic reaction.

$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

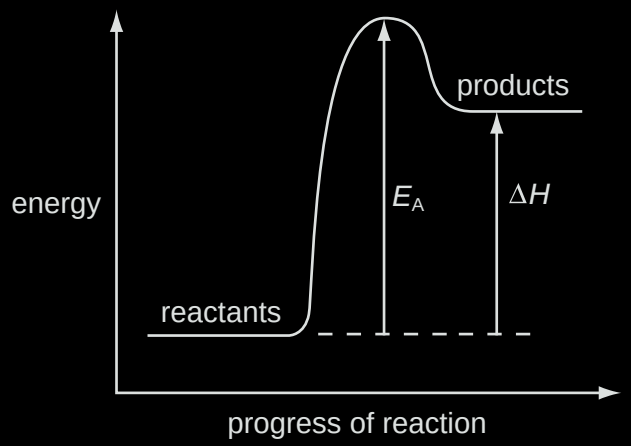
Which statement about this exothermic reaction is **not** correct?

A Magnesium chloride is soluble in water.

B Magnesium is above hydrogen in the reactivity series.

C One mole of magnesium produces one mole of hydrogen gas.

D The total energy of the products is greater than that of the reactants.



What is the correct description of the reaction?

	sign of ΔH	overall energy change	sign of E_A
A	–	exothermic	–
B	+	endothermic	+
C	+	endothermic	–
D	+	exothermic	+

17 In the Contact process for making sulphuric acid, one step involves the oxidation of sulphur dioxide as shown below.

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

The forward reaction is exothermic.

Which change would increase the amount of sulphur trioxide produced at equilibrium?

A increasing the temperature

B decreasing the temperature

C decreasing the pressure

D adding a catalyst

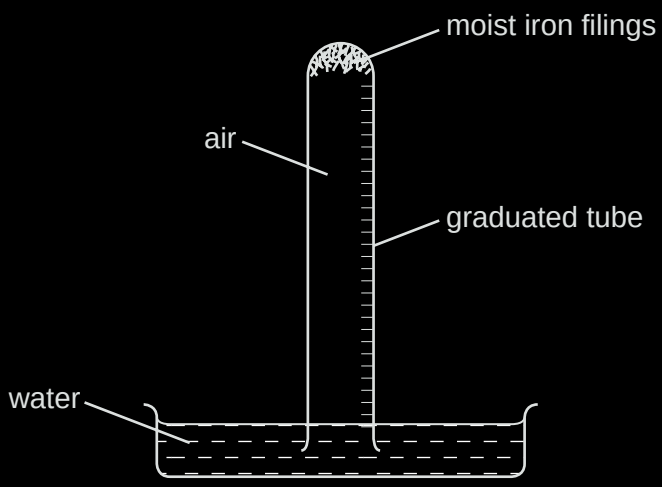
18 Which statement about conduction of electricity is correct?

A Electricity is conducted in aqueous solution by electrons.

B Electricity is conducted in a metal wire by ions.

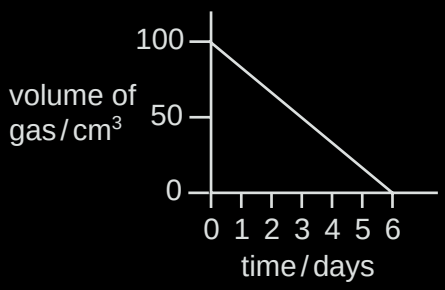
C Electricity is conducted in a molten electrolyte by electrons.

D Electricity is conducted in an acid solution by ions.

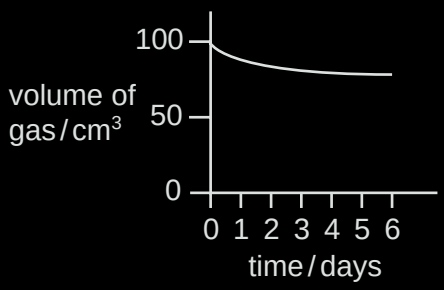


Which graph best represents how the volume of gas changes with time?

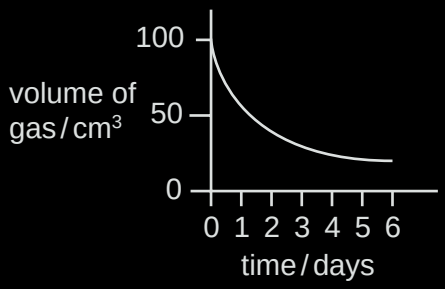
A



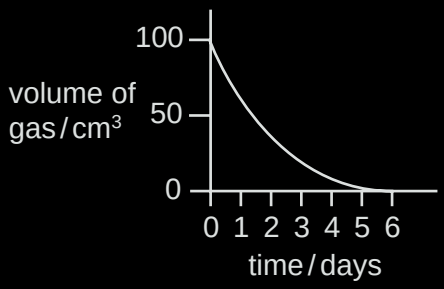
B



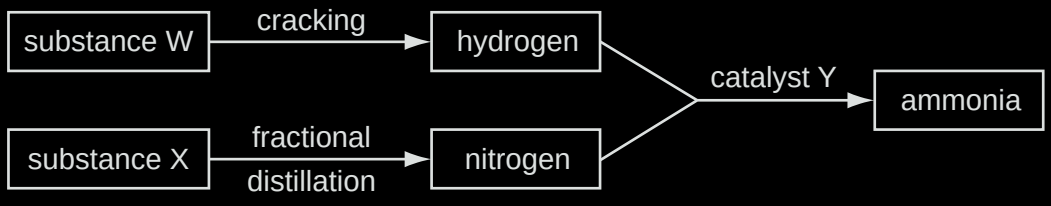
C



D







What are substances W and X and catalyst Y?

	W	X	Y
A	air	oil	iron
B	air	oil	vanadium(V) oxide
C	oil	air	iron
D	oil	air	vanadium(V) oxide

33 Element **R** reacts with oxygen to form a gas, **T**.

T changes the colour of damp litmus paper from blue to red.

T is used to kill bacteria in the preservation of dried fruit.

What is **R**?

A carbon

B chlorine

C nitrogen

D sulphur

34 The gases coming from a car's exhaust contain oxides of nitrogen.

How are these oxides formed?

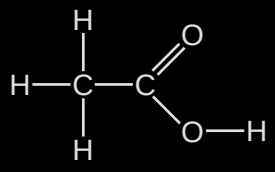
A Nitrogen reacts with carbon dioxide.

B Nitrogen reacts with carbon monoxide.

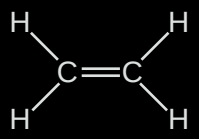
C Nitrogen reacts with oxygen.

D Nitrogen reacts with petrol.

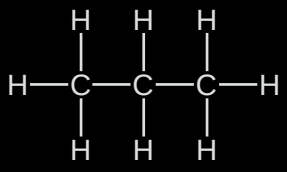
1



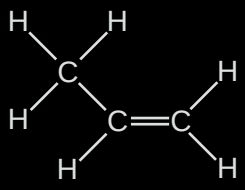
2



3



4



Which compounds decolourise bromine water?

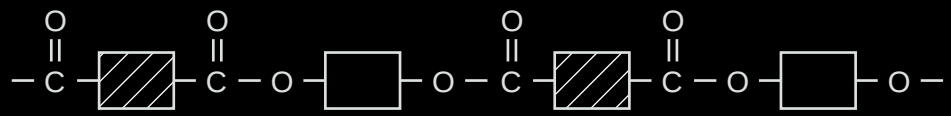
A 1 and 2 **B** 2 and 4 **C** 3 only **D** 3 and 4

40 Which polymer would hydrolyse to amino acids?

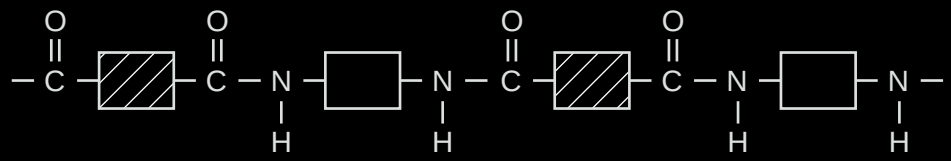
A



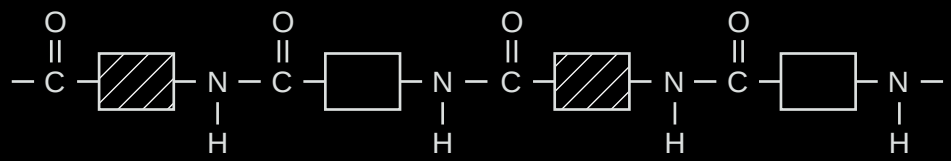
B



C



D



DATA SHEET
The Periodic Table of the Elements

Group																
I	II										III	IV	V	VI	VII	0
							1 H Hydrogen 1									4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4										11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12										27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulphur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	64 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
85 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57 *	178 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	Po Polonium 84	At Astatine 85	Rn Radon 86
Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89 †														

140 Ce Cerium 58	141 Pr Praseodymium 59	144 Nd Neodymium 60	Pm Promethium 61	150 Sm Samarium 62	152 Eu Europium 63	157 Gd Gadolinium 64	159 Tb Terbium 65	162 Dy Dysprosium 66	165 Ho Holmium 67	167 Er Erbium 68	169 Tm Thulium 69	173 Yb Ytterbium 70	175 Lu Lutetium 71
232 Th Thorium 90	Pa Protactinium 91	238 U Uranium 92	Np Neptunium 93	Pu Plutonium 94	Am Americium 95	Cm Curium 96	Bk Berkelium 97	Cf Californium 98	Es Einsteinium 99	Fm Fermium 100	Md Mendelevium 101	No Nobelium 102	Lr Lawrencium 103

*58-71 Lanthanoid series
†90-103 Actinoid series

Key

a
X
b

a = relative atomic mass
X = atomic symbol
b = proton (atomic) number

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).